

Global AI Startup Funding Trend Analysis

(Sector / Year-wise)

1. Introduction

Artificial Intelligence (AI) has become an important technology in today's business and technology environment. AI is being used in many areas such as healthcare, finance, education, cybersecurity, robotics, transportation, manufacturing, and Generative AI.

The growth of AI has also resulted in the growth of AI startups. These startups require funding for research and development, product development, hiring employees, infrastructure, marketing, and business expansion.

This research paper focuses on **Global AI Startup Funding Trend Analysis**. The project analyzes AI startup funding based on different sectors, years, countries, and funding stages.

The main objective of this project is to understand how AI startup funding changes over time and how funding is distributed across different AI sectors, countries, and investment stages.

2. Growth of Artificial Intelligence Startups

Artificial Intelligence has developed rapidly because of improvements in computing power, cloud technology, availability of data, machine learning algorithms, and digital infrastructure.

AI startups are working in several areas, including:

- Generative AI
- Cybersecurity AI
- Healthcare AI
- Robotics
- FinTech AI
- EdTech AI
- Autonomous Vehicles
- Machine Learning
- AI Software
- Data and Analytics

The growth of Generative AI has also increased interest in AI-based products and services. As the AI industry grows, startups require more financial support to develop new technologies and expand their businesses.

Therefore, studying startup funding provides useful information about the growth and development of the AI startup ecosystem.

3. Global AI Startup Funding

AI startup funding includes investments provided by venture capital firms, companies, investors, and other financial organizations.

Startup funding can be used for:

- Research and development
- Product development
- Hiring employees
- Technology infrastructure
- Cloud computing
- Marketing
- Business expansion
- AI research
- Development of new AI products

Funding can vary depending on the startup's sector, country, development stage, and business model.

This project analyzes these funding patterns using data analytics techniques.

The dataset used in this project contains **1,220 raw records**. After data cleaning and processing, **1,201 records** are available for analysis.

The dataset covers:

- **16 countries**
 - **15 AI sectors**
 - **7 funding stages**
 - **2018 to 2025**
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4. AI Startup Funding Stages

AI startups receive funding at different stages of their development.

The funding stages included in this project are:

- Pre-Seed
- Seed
- Series A
- Series B
- Series C
- Series D
- Late Stage

Early-stage funding generally supports the initial development of a startup, while later-stage funding can support expansion and larger business operations.

The project analyzes funding at each stage to understand how the total funding is distributed across the startup lifecycle.

5. Sector-wise AI Funding

AI startups operate in different sectors. Different sectors can attract different amounts of investment depending on market demand, technology development, and business opportunities.

The project includes sectors such as:

- Cybersecurity AI
- Generative AI
- EdTech AI
- Autonomous Vehicles
- Robotics
- Healthcare AI
- FinTech AI
- Other AI-related sectors

Sector-wise analysis helps identify how funding is distributed among different areas of AI.

Based on the cleaned project dataset, some of the major sectors by recorded funding are:

- Cybersecurity AI – approximately USD 3,072.40 million
- Generative AI – approximately USD 3,061.21 million
- EdTech AI – approximately USD 2,995.78 million
- Autonomous Vehicles – approximately USD 2,663.46 million
- Robotics – approximately USD 2,500.01 million

These values represent the project dataset and are used for analytical comparison.

6. Year-wise AI Startup Funding

The project analyzes AI startup funding from **2018 to 2025**.

The total recorded funding for each year in the cleaned dataset is:

Year Total Funding (USD Million)	
2018	2,895.57
2019	2,792.14
2020	3,414.88
2021	4,323.27
2022	3,772.38

Year Total Funding (USD Million)

2023	5,175.73
2024	5,112.47
2025	6,521.23

The project dataset records its highest yearly funding in **2025**, with approximately **USD 6,521.23 million**.

Year-wise analysis helps to understand changes in AI startup funding over time.

7. Country-wise AI Startup Funding

The dataset contains AI startup information from 16 countries.

Country-wise analysis is performed using total funding and startup information.

Some of the countries with higher recorded funding in the project dataset are:

- Netherlands – USD 2,927.06 million
- Singapore – USD 2,851.92 million
- Japan – USD 2,772.78 million
- South Korea – USD 2,704.74 million
- Canada – USD 2,425.64 million

These values are based only on the records included in the project dataset. They should not be considered as official national AI investment statistics.

Country-wise analysis helps to understand the geographical distribution of AI startup funding.

8. Funding Stage Analysis

The project also analyzes total funding according to funding stage.

The recorded funding by stage is:

Funding Stage Total Funding (USD Million)

Pre-Seed	63.62
Seed	647.96
Series A	1,909.78
Series B	3,906.50
Series C	4,760.66
Series D	8,027.10

Funding Stage Total Funding (USD Million)

Late Stage	14,692.05
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The data shows that later funding stages represent a large portion of the recorded funding in this dataset.

This analysis helps understand how funding is distributed between early-stage and later-stage AI startups.

9. Startup Valuation and Funding

The dataset also contains startup valuation information.

Valuation represents the estimated value of a startup at a particular stage. Comparing valuation with funding can provide additional information about the startup's financial position.

The project creates a derived variable called:

Valuation-to-Funding Ratio

Formula:

Valuation-to-Funding Ratio = Valuation / Funding Amount

This measure is used as an additional analytical feature for comparing startups.

10. Data Analysis Methodology

The project follows an end-to-end data analytics workflow:

Raw Dataset

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Data Cleaning

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Feature Engineering

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Exploratory Data Analysis

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MySQL Database

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SQL Analysis

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Python Analysis

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Matplotlib + Seaborn

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Power BI Dashboard

The raw CSV dataset is first loaded using Python and Pandas.

The data is then cleaned by checking missing values, duplicate records, incorrect data types, invalid values, and inconsistent sector names.

After cleaning, additional features are created.

The cleaned data is stored in MySQL for structured database analysis.

SQL queries are then used for sector-wise, year-wise, country-wise, and funding-stage analysis.

Python is used for exploratory data analysis and visualization.

Finally, Power BI is used to present the results through an interactive dashboard.

11. Data Cleaning

Data cleaning is performed before the main analysis.

The following operations are performed:

- Checking dataset structure
- Checking missing values
- Checking duplicate records
- Removing duplicate records
- Converting numerical columns
- Checking invalid funding values

- Handling missing values
- Standardizing sector names
- Checking country names
- Checking funding stages

The raw dataset contains **1,220 records**.

After the cleaning process, the final dataset contains **1,201 records**.

12. Feature Engineering

Feature engineering is used to create additional columns that help in analysis.

The project creates the following features:

1. Funding Size

Funding amount is converted into different funding-size categories.

2. Funding Per Employee

This shows funding in relation to the number of employees.

Formula:

Funding Per Employee = Funding Amount / Employees

3. Valuation to Funding Ratio

This compares startup valuation with its funding amount.

Formula:

Valuation to Funding Ratio = Valuation / Funding Amount

4. Funding Per Investor

This calculates funding amount in relation to the number of investors.

Formula:

Funding Per Investor = Funding Amount / Investors Count

These features provide additional information for the analysis.

13. SQL and MySQL Analysis

MySQL is used to store the cleaned AI startup funding data.

The database used in the project is:

Database: `global_ai_funding`

Table: `global_ai_startup_funding`

SQL is used to perform different types of analysis.

The project uses SQL queries for:

- Total record count
- Available years
- Total funding
- Average funding
- Maximum funding
- Total valuation
- Sector-wise funding
- Sector-wise startup count
- Year-wise funding
- Year-wise startup count
- Country-wise funding
- Funding-stage analysis
- Top-funded startups
- Top countries
- Sector-year analysis

SQL helps perform analysis directly from the database.

14. Python and MySQL Connectivity

Python is connected with MySQL using the **MySQL Connector/Python** library.

The Python program connects to:

Database: `global_ai_funding`

Table: `global_ai_startup_funding`

Python sends SQL queries to MySQL and retrieves the required results.

This connection allows the project to combine Python-based data analysis with database-level SQL analysis.

15. Data Visualization

Data visualization is an important part of the project.

Matplotlib and Seaborn are used to create different charts.

The project includes analysis such as:

- Year-wise funding trend
- Sector-wise funding
- Country-wise funding
- Funding-stage distribution
- Startup count by year
- Sector-wise startup count
- Sector and year comparison

Visualization makes it easier to understand large datasets and identify trends.

16. Power BI Dashboard

Power BI is used to create an interactive dashboard for the project.

The dashboard can display important information through:

- KPI Cards
- Bar Charts
- Line Charts
- Donut Charts
- Tables
- Slicers
- Filters

Important dashboard metrics include:

- Total Funding
- Total Startups
- Average Funding
- Total Valuation
- Total Employees
- Total Investors

The dashboard allows users to filter the data by:

- Year
- Country
- Sector

- Funding Stage

This makes the analysis easier to explore and understand.

17. Results

The analysis of the cleaned dataset provides the following results:

- Total records: **1,201**
- Countries: **16**
- AI sectors: **15**
- Funding stages: **7**
- Analysis period: **2018–2025**
- Total recorded funding: approximately **USD 34,007.67 million**
- Average funding: approximately **USD 28.32 million**
- Maximum individual funding: approximately **USD 732.34 million**

The year-wise analysis shows that 2025 has the highest recorded annual funding in the dataset.

The sector-wise analysis shows different funding levels across AI application areas.

The funding-stage analysis shows that later-stage funding accounts for a large portion of the total recorded funding.

18. Discussion

The analysis shows that AI startup funding changes across different years, sectors, countries, and funding stages.

The year-wise analysis helps understand how funding changes over time. Sector-wise analysis shows that investment is distributed across different AI application areas.

Country-wise analysis provides a geographical view of the startup funding ecosystem.

Funding-stage analysis shows that the amount of funding can increase substantially as startups move towards later investment stages.

The project therefore provides multiple views of the global AI startup funding ecosystem instead of analyzing only one variable.

19. Limitations

The project has some limitations:

- The dataset does not necessarily contain every AI startup in the world.
 - The analysis depends on the records available in the project dataset.
 - Funding values may vary depending on the source and reporting method.
 - Country-wise values should not be treated as official national investment statistics.
 - The project mainly focuses on descriptive analysis.
 - The analysis does not prove that one factor directly causes another.
 - External factors such as market conditions and economic changes are not included in the dataset.
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20. Future Trends

AI startup funding is expected to remain an important area of the technology industry.

Future investment may focus on areas such as:

- Generative AI
- AI Infrastructure
- Data Centers
- AI Chips and Semiconductors
- Robotics
- Autonomous Systems
- Healthcare AI
- Cybersecurity AI
- AI Software
- Cloud Computing
- AI Research
- AI Education and Skills

As AI technologies continue to develop, startup funding can also expand into new application areas.

Future versions of this project can include more recent funding records and additional countries and sectors.

21. Future Scope of the Project

The project can be extended in the future by:

1. Adding more recent AI startup funding data.
2. Adding more countries.
3. Adding investor-level analysis.
4. Adding startup acquisition and IPO information.
5. Performing funding prediction using machine learning.
6. Performing time-series forecasting.

7. Adding employee growth analysis.
 8. Adding automated data updates.
 9. Connecting Power BI directly with MySQL.
 10. Adding real-time or scheduled dashboard refresh.
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22. Conclusion

Artificial Intelligence has become an important area of technology and business development. The growth of AI startups has created a large requirement for funding and investment.

This project, **Global AI Startup Funding Trend Analysis (Sector/Year-wise)**, analyzes AI startup funding using different dimensions such as year, sector, country, and funding stage.

The project demonstrates how raw startup funding data can be converted into useful information using data cleaning, feature engineering, exploratory data analysis, SQL queries, Python visualization, and Power BI dashboards.

The project uses **Python, Pandas, NumPy, MySQL, SQL, MySQL Connector/Python, Matplotlib, Seaborn, Jupyter Notebook, and Power BI** as the main technologies.

The final analysis provides a clear view of how AI startup funding is distributed across different sectors, countries, years, and funding stages.

Overall, the project demonstrates a complete practical **Data Analytics workflow**, starting from raw data and ending with analysis, visualization, and an interactive dashboard.

23. References

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